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Pada soal pertama kami diberi soal seperti ini

Wireshark twoo twooo two twoo...

Medium

Forensics

picoCTF 2021

AUTHOR: DYLAN

## Description

Can you find the flag? [shark2.pcapng](#).

Sama seperti sebelumnya, kami diberi file yang bisa dibuka di wireshark.

|    |          |                |                |         |      |                                                           |  |
|----|----------|----------------|----------------|---------|------|-----------------------------------------------------------|--|
| 1  | 0.000000 | 192.168.38.104 | 192.168.38.105 | TCP     | 54   | 63631 → 8412 [FIN, ACK] Seq=1 Ack=1 Win=8394 Len=0        |  |
| 2  | 0.000380 | 192.168.38.105 | 192.168.38.104 | TLSv1.2 | 78   | Application Data                                          |  |
| 3  | 0.000430 | 192.168.38.104 | 192.168.38.105 | TCP     | 54   | 63631 → 8412 [RST, ACK] Seq=2 Ack=25 Win=0 Len=0          |  |
| 4  | 0.000458 | 192.168.38.105 | 192.168.38.104 | TCP     | 54   | 8412 → 63631 [FIN, ACK] Seq=25 Ack=2 Win=227 Len=0        |  |
| 5  | 0.000636 | 192.168.38.104 | 192.168.38.105 | TCP     | 66   | 63633 → 8412 [SYN] Seq=0 Win=62727 Len=0 MSS=8961 WS=256  |  |
| 6  | 0.003348 | 192.168.38.105 | 192.168.38.104 | TCP     | 66   | 8412 → 63633 [SYN, ACK] Seq=0 Ack=1 Win=26883 Len=0 MSS=8 |  |
| 7  | 0.003403 | 192.168.38.104 | 192.168.38.105 | TCP     | 54   | 63633 → 8412 [ACK] Seq=1 Ack=1 Win=2150400 Len=0          |  |
| 8  | 0.004374 | 192.168.38.104 | 192.168.38.105 | TLSv1.3 | 571  | Client Hello (SNI=kolide)                                 |  |
| 9  | 0.004602 | 192.168.38.105 | 192.168.38.104 | TCP     | 54   | 8412 → 63633 [ACK] Seq=1 Ack=518 Win=28032 Len=0          |  |
| 10 | 0.007436 | 192.168.38.105 | 192.168.38.104 | TLSv1.3 | 1420 | Server Hello, Change Cipher Spec, Application Data, Appl  |  |
| 11 | 0.008274 | 192.168.38.104 | 192.168.38.105 | TLSv1.3 | 118  | Change Cipher Spec, Application Data                      |  |
| 12 | 0.049378 | 192.168.38.105 | 192.168.38.104 | TCP     | 54   | 8412 → 63633 [ACK] Seq=1367 Ack=582 Win=28032 Len=0       |  |
| 13 | 0.049431 | 192.168.38.104 | 192.168.38.105 | TLSv1.3 | 495  | Application Data, Application Data, Application Data, App |  |
| 14 | 0.049684 | 192.168.38.105 | 192.168.38.104 | TCP     | 54   | 8412 → 63633 [ACK] Seq=1367 Ack=1023 Win=29056 Len=0      |  |
| 15 | 0.053117 | 192.168.38.105 | 192.168.38.104 | TLSv1.3 | 219  | Application Data                                          |  |
| 16 | 0.093304 | 192.168.38.104 | 192.168.38.105 | TCP     | 54   | 63633 → 8412 [ACK] Seq=1023 Ack=1532 Win=2148864 Len=0    |  |

  

|                                          |                                                              |      |                                                 |         |
|------------------------------------------|--------------------------------------------------------------|------|-------------------------------------------------|---------|
| Frame 1:                                 | 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on | 0000 | 02 4d 59 32 10 a1 02 3b c6 1a ae f5 08 00 45 00 | MY2...  |
| Ethernet II, Src:                        | MS-NLB-PhysServer-32_1b:c6:1a:ae:f5 (02:3b:c6:1a:ae:f5)      | 0010 | 00 28 49 ed 40 00 80 06 00 00 c0 a8 26 68 c0 a8 | (I @... |
| Internet Protocol Version 4, Src:        | 192.168.38.104, Dst: 192.168.38.105                          | 0020 | 26 69 f8 8f 20 dc ee 51 1a ba 23 f2 9f 55 50 11 | &i...Q  |
| Transmission Control Protocol, Src Port: | 63631, Dst Port: 8412, Seq: 1                                | 0030 | 20 ca ce 3c 00 00                               | ...<... |

Setelah itu, kita ke halaman statistic dan kemudian ke protocol hierarchy dan kita apply filter DNS atau “Domain Name System”

|     |          |                |                |     |     |                         |                                                         |  |
|-----|----------|----------------|----------------|-----|-----|-------------------------|---------------------------------------------------------|--|
| 791 | 7.931626 | 192.168.38.104 | 8.8.8.8        | DNS | 93  | Standard query          | 0x76aa A lDqoR16q.reddshrimpandherring.com              |  |
| 792 | 7.943025 | 8.8.8.8        | 192.168.38.104 | DNS | 166 | Standard query response | 0x76aa No such name A lDqoR16q.reddshrimpandherring.com |  |
| 793 | 7.947216 | 192.168.38.104 | 8.8.8.8        | DNS | 131 | Standard query          | 0xcdd5 A lDqoR16q.reddshrimpandherring.com              |  |
| 794 | 7.957680 | 8.8.8.8        | 192.168.38.104 | DNS | 203 | Standard query response | 0xcdd5 No such name A lDqoR16q.reddshrimpandherring.com |  |
| 795 | 7.958549 | 192.168.38.104 | 8.8.8.8        | DNS | 109 | Standard query          | 0x5d2d A lDqoR16q.reddshrimpandherring.com              |  |
| 796 | 7.967998 | 8.8.8.8        | 192.168.38.104 | DNS | 184 | Standard query response | 0x5d2d No such name A lDqoR16q.reddshrimpandherring.com |  |
| 797 | 7.968981 | 192.168.38.104 | 8.8.8.8        | DNS | 93  | Standard query          | 0xc847 A 1Th0dQuT.reddshrimpandherring.com              |  |
| 798 | 8.049550 | 8.8.8.8        | 192.168.38.104 | DNS | 166 | Standard query response | 0xc847 No such name A 1Th0dQuT.reddshrimpandherring.com |  |
| 799 | 8.050527 | 192.168.38.104 | 8.8.8.8        | DNS | 131 | Standard query          | 0x21a5 A 1Th0dQuT.reddshrimpandherring.com              |  |
| 800 | 8.061483 | 8.8.8.8        | 192.168.38.104 | DNS | 205 | Standard query response | 0x21a5 No such name A 1Th0dQuT.reddshrimpandherring.com |  |
| 801 | 8.062385 | 192.168.38.104 | 8.8.8.8        | DNS | 109 | Standard query          | 0xc006 A 1Th0dQuT.reddshrimpandherring.com              |  |
| 802 | 8.071914 | 8.8.8.8        | 192.168.38.104 | DNS | 184 | Standard query response | 0xc006 No such name A 1Th0dQuT.reddshrimpandherring.com |  |
| 803 | 8.073001 | 192.168.38.104 | 8.8.8.8        | DNS | 93  | Standard query          | 0x4249 A ysnuBebx.reddshrimpandherring.com              |  |
| 804 | 8.142293 | 8.8.8.8        | 192.168.38.104 | DNS | 166 | Standard query response | 0x4249 No such name A ysnuBebx.reddshrimpandherring.com |  |
| 805 | 8.143556 | 192.168.38.104 | 8.8.8.8        | DNS | 131 | Standard query          | 0xb326 A ysnuBebx.reddshrimpandherring.com              |  |
| 806 | 8.155877 | 8.8.8.8        | 192.168.38.104 | DNS | 203 | Standard query response | 0xb326 No such name A ysnuBebx.reddshrimpandherring.com |  |

  

|                                                                                                                                                                                                                                                                                                                                         |  |  |  |                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Frame 791: 93 bytes on wire (744 bits), 93 bytes captured (744 bits) on interface 0<br>Ethernet II, Src: MS-NLB-PhysServer-32 1b:c6:1a:ae:f5 (02:3b:c6:1a:ae:f5), Dst: 8.8.8.8<br>Internet Protocol Version 4, Src: 192.168.38.104, Dst: 8.8.8.8<br>User Datagram Protocol, Src Port: 65299, Dst Port: 53<br>Domain Name System (query) |  |  |  | 0000 02 fb 68 4c e9 41 02 3b c6 1a ae f5 08 00 45 00 hL A ;<br>0010 00 4f 73 e3 00 00 80 11 00 00 c0 a8 26 68 08 08 Os<br>0020 08 08 ff 13 00 35 00 3b f7 6c 76 aa 01 00 00 01 5 ; lv<br>0030 00 00 00 00 00 00 08 6c 44 71 6f 52 31 36 71 14 l Dqo<br>0040 72 65 64 64 73 68 72 69 6d 70 61 6e 64 68 65 72 reddshri mpa<br>0050 72 69 6e 67 03 63 6f 6d 00 00 01 00 01 ring.com |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Kemudian, kita harus mengecek satu persatu pada paket yang dikirim pada “Line-based text”, dan juga mengecek IP yang terlihat mencurigakan. Setelah itu, kita pasang filter untuk sorting log yang mencurigakan yaitu dengan line “dns && ip.dst == 18.217.1.57 && dns.qry.name contains local”

|      |           |                |             |     |     |                |                                                           |  |
|------|-----------|----------------|-------------|-----|-----|----------------|-----------------------------------------------------------|--|
| 1637 | 9.440363  | 192.168.38.104 | 18.217.1.57 | DNS | 109 | Standard query | 0x1dd2 A cG1jb0NURntkbnNfM3hmMWxfZnR3X2RlYWRiZWVmfQ=      |  |
| 2046 | 11.972605 | 192.168.38.104 | 18.217.1.57 | DNS | 109 | Standard query | 0xab9 A RntkbnNf.reddshrimpandherring.com.win             |  |
| 2448 | 14.605726 | 192.168.38.104 | 18.217.1.57 | DNS | 109 | Standard query | 0x9e21 A M3hmMWxf.reddshrimpandherring.com.win            |  |
| 3153 | 16.506492 | 192.168.38.104 | 18.217.1.57 | DNS | 109 | Standard query | 0x2ee1 A ZnR3X2RlYWRiZWVmfQ=.reddshrimpandherring.com.win |  |
| 3442 | 18.340155 | 192.168.38.104 | 18.217.1.57 | DNS | 109 | Standard query | 0x2a4b A YWRiZWVmfQ=.reddshrimpandherring.com.win         |  |
| 3982 | 20.369626 | 192.168.38.104 | 18.217.1.57 | DNS | 105 | Standard query | 0x4068 A fQ=.reddshrimpandherring.com.window              |  |
| 4374 | 22.583745 | 192.168.38.104 | 18.217.1.57 | DNS | 105 | Standard query | 0x7418 A fQ=.reddshrimpandherring.com.window              |  |

Pada baris 1-4 terlihat sebuah perbedaan dibanding line yang lain dan terlihat mencurigakan sehingga saya mencoba mendecode 4 string diatas yang digabungkan menjadi 1 sehingga menjadi seperti ini  
 “cG1jb0NURntkbnNfM3hmMWxfZnR3X2RlYWRiZWVmfQ=”

Disini dapat disimpulkan jika decode yang digunakan adalah base64 karena sesuai ciri-cirinya hanya terdiri a-z, A-Z, +, =, -, /.

Recipe

From Base64

Alphabet  
A-Za-z0-9+/=

☒ Remove non-alphabet chars

☐ Strict mode

Input

cGljb0NURntkbnNfM3hmMlwxZnR3X2RlYWRIZWVmfQ=

REC 43 1 0~42 (42 selected)

Output

picoCTF{dns\_3xf1l\_ftw\_deadbeef}

Sehingga kita menemukan flagnya.